

HOUSE PRICE PREDICTION

DECLARATION

I, P. YEMEEMA , hereby declare that this project report, titled " **HOUSE PRICE PREDICTION** ", is the result of my original work completed during the 6-month IT/Programming Language training course at Vihara Tech. This project was executed over the full term, incorporating the practical knowledge gained during the four months of intensive classroom training and the real-world experience acquired during the subsequent two-month paid internship at The Skill Union. The information and findings presented herein are a true and accurate reflection of the work performed and have not been submitted, in whole or in part, for any other degree or diploma. I confirm that all sources of information have been specifically acknowledged.

Name

Batch / Course

Signature

P. YEMEEMA

Data Science

ACKNOWLEDGEMENT

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1.Acknowledgements

Dataset Source

- **Dataset Name:** House Sales in King County, USA.
- **Source:** Kaggle - House Data dataset.
- **Contributor:** Shreya Chaudhary.
- **Context:** This dataset contains house sale prices for King County, including Seattle, sold between May 2014 and May 2015.

Tools and Libraries

- **Flask:** Web framework for Python.
- **Scikit-learn:** Used for train-test splitting.
- **NumPy:** Used for custom numerical computations and SVD matrix operations.
- **Pickle:** Model serialization for the MINI_PROJECT.pkl file.
- **Pandas:** Data loading and preprocessing.
- **HTML/CSS:** Frontend development for the user interface.
- **PyCharm:** Primary IDE for professional Python development and environment management.

2.Project Overview

Objective:

The primary goal is to develop a machine learning web application using a custom Linear Regression implementation to predict house prices based on features like location, size, and condition.

Problem Statement:

Real estate prices are complex to analyse manually due to multiple influencing factors. This project aims to:

- Create a predictive model using historical data without relying on standard library regression models.
- Build a responsive web interface for instant price estimation.
- Deploy the application for public access via the Render platform

3.Dataset Description

Specifications

- Total Records: 4800 houses.
- Features: 18 initial columns, with 17 features selected for the final model.
- Location: King County, Washington, USA

Data Preprocessing Steps

1. **Handling Missing Values:** Checked for null values to ensure data quality.
2. **Date Feature Engineering:** Extracted year, month, and day from the original date column.
3. **Encoding Categorical Variables:** Manual Label Encoding was applied to the city column, and the country column was mapped to numeric values.

4. Project Structure

The project is organized in **PyCharm** as follows:

- **MINI_PROJECT/** (Root Directory)
 - **.venv/**: Virtual environment library root.
 - **static/**: Contains static assets.
 - **Images/**: Hema.jpeg for the UI.
 - **templates/**: Contains index.html for the web frontend.
 - **app.py**: The main Flask application script.
 - **data.csv**: The source house sales dataset.
 - **MINI_PROJECT.pkl**: The serialized custom-trained model.
 - **Procfile**: Configuration for production deployment.
 - **requirements.txt**: List of all Python dependencies

5. Technology Stack

- **Backend**: Python and Flask.
- **Numerical Core**: NumPy and Pandas.
- **Frontend**: HTML5 and CSS3 with modern animations.
- **IDE**: PyCharm for coding and version control integration.
- **Deployment**: Render (Cloud Hosting) and GitHub (Version Control)

6. Implementation Steps

Step 1: Environment Setup in PyCharm

- Create and activate a virtual environment (venv) to isolate project packages.
- Install dependencies: flask, numpy, pandas, scikit-learn, and gunicorn.

Step 2: Custom Model Training

- Load data and perform EDA.
- Implement the MINIPROJECT class for custom Linear Regression using **Singular Value Decomposition (SVD)**.
- Save the resulting weights and intercept into MINI_PROJECT.pkl using Pickle.

Step 3: Web Application & Deployment

- Develop Flask routes to handle input from the 17-feature form.
- Use GitHub to host the repository.
- Connect GitHub to Render for automatic deployment via the Procfile.

7. Code Explanation

Class: MINIPROJECT This class encapsulates the ML workflow:

- **__init__**: Loads the CSV and performs feature engineering on dates and categorical cities.
- **training()**: Implements Linear Regression using **Singular Value Decomposition (SVD)**.
 - **Mean Centering**: Centers features and targets to improve numerical stability.

- **SVD Application:** Computes the pseudo-inverse of the matrix using `np.linalg.svd`.
- **testing():** Calculates predictions on unseen data using the calculated coefficients (`$m$`) and intercept (`$c$`)

8. Model Development

Why Custom Linear Regression?

- **Interpretability:** Provides clear coefficients for each feature.
- **Mathematical Precision:** Using SVD allows for solving the normal equation even if the feature matrix is rank-deficient.
- **Efficiency:** Low computational cost for real-time web predictions.

Performance Metrics

- **RMSE:** Measures the average deviation between predicted and actual house prices.
- **R² Score:** Indicates how much of the price variance is successfully captured by the model.

9. Deployment Process

The project uses a **CI/CD pipeline** via GitHub and Render:

1. **GitHub Upload:** All project files, including the `.pkl` model, are uploaded to a repository.
2. **Render Configuration:** A "Web Service" is created on Render pointing to the GitHub URL.
3. **Build & Start:** Render installs dependencies from `requirements.txt` and runs the app using `gunicorn app:app`

10. Results and Output

- **User Interface:** A clean form with 17 fields for inputs like sqft_living, condition, and year built.
- **Sample Result:** For a specific set of house features (e.g., 4 bedrooms, 4 bathrooms, 1340 sqft), the model estimated a price of approximately **7.688e+08**.
- **Response Time:** Predictions are returned in less than 2 seconds

11. Challenges and Solutions

- **Feature Encoding:** Categorical city names were manually mapped to integers to be processed by the SVD algorithm.
- **Model Persistence:** The custom class state had to be serialized into a .pkl file for the Flask app to load without retraining.
- **Data Types:** Form inputs arrive as strings; the solution was to convert all inputs to floats using a list comprehension: `[float(i) for i in request.form.values()]`

12. Future Enhancements

- **Algorithms:** Implementing Random Forest or Gradient Boosting to compare against the linear baseline.
- **UI Features:** Adding historical price trend charts and data visualization.
- **Scalability:** Containerizing the application using **Docker** for more consistent deployments.

13. Conclusion

The House Price Prediction project successfully demonstrates a full machine learning pipeline. By implementing a custom Linear Regression model and deploying it via Flask and Render, it provides a practical tool for real estate estimation and serves as a robust template for future ML web applications.

14. References

1. **Flask Documentation:** <https://flask.palletsprojects.com/>
2. **NumPy SVD Guide:**
<https://numpy.org/doc/stable/reference/generated/numpy.linalg.svd.html>
3. **Render Deployment:** <https://house-price-predictions1.onrender.com>
4. **Dataset Source:** Kaggle - House Sales in King County, USA
5. **Github repositories :** <https://github.com/pamulayemeema/House-Price-Prediction>